



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

**COURSE: ADAPTIVE SOFTWARE ENGINEERING COURSE CODE : 24CI3201 A.Y.
2026 - 2027**

TITLE OF THE PROJECT	SmartInvest – AI-Powered Multi-Asset Investment and Margin Trading Platform	
S. No.	University ID	Name
1	2420030732	K. Bodhi Sathwik
2	2420030720	Badri Ramagiri
3	2420030725	Nandhu
Name of the Guide		Dr. Archana Kalidindi

Abstract

SmartInvest is a full-stack, AI-powered multi-asset investment and portfolio management platform developed using modern **Software Engineering** and **Agile Development** practices. The system provides a unified environment for managing stocks, mutual funds, Futures & Options (F&O), portfolio tracking, transactions, profit and loss, asset allocation, performance analytics, margin management, and collateral pledging. The project follows the **Software Development Life Cycle (SDLC)** using an **Agile methodology** and **Scrum framework**, with **User Stories**, **Product Backlog**, **Sprint Planning**, **Backlog Refinement**, **Estimation**, **Velocity**, and continuous team communication to support iterative development. **Git** is used for version control and collaborative development, while **CI/CD** practices are incorporated to automate the build, testing, and deployment process. The application follows a **Cloud-Native** architecture and can be containerized using **Docker**, with **Kubernetes** and **Minikube** providing orchestration and deployment capabilities. Security is integrated through **DevSecOps** practices, including **Secure Coding**, authentication using JWT and bcrypt, **Static Code Analysis** using **SonarQube**, **Vulnerability Scanning** using **Trivy**, and security testing based on the **OWASP Top 10** principles. The platform incorporates an AI-powered Investment Analyst using the Google Gemini API to analyze portfolio diversification, risk exposure, asset concentration, margin utilization, and overall portfolio health. The AI component follows **MLOps** concepts such as **Data Ingestion**, **Model Training**, **Model Versioning**, **Model Deployment**, and **Model Monitoring**, with **DVC** and **MLflow** supporting data/model versioning and experiment tracking. **Prometheus** and **Grafana** can be used for application and model **Observability** and monitoring. **Cloud Deployment** using **AWS/GCP** enables scalable storage, APIs, and deployment. The project also considers **Bias and Fairness in AI**, **Ethical Considerations**, and secure handling of user data. Overall, SmartInvest demonstrates the practical application of **Agile**, **DevOps**, **DevSecOps**, **MLOps**, **CI/CD**, **Cloud Computing**, **Security**, **Observability**, and **AI** concepts to develop a secure, scalable, and intelligent investment management platform.

Signature of the Guide

HOD